

Applications of a spatial analysis system to ERM losses management

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Abstract

This paper provides description of a system using statistical spatial methods to measuring, analysis, and managing ERM losses. An example of its application to a delinquency rate in a residential mortgage portfolio analysis demonstrates how this methodology can help decreasing the company's exposure in high risk areas, and increasing exposure in the low risk areas.

Introduction

The Casualty Actuarial Society (CAS (2003)) classified the Enterprise Risk Management (ERM) risks in the following four groups: 1) hazard risks (liability torts, property damage, and natural catastrophe), 2) financial risks (pricing risk, asset risk, currency risk, liquidity risk); 3) operational risk (customer satisfaction, product failure, integrity, reputational risk); 4) strategic risks (competition, social trend, capital availability).

A spatial analysis methodology suggested below addresses the key responsibilities any ERM analyst faces: a) evaluating the risks, b) developing a distributional analysis that allows comparing the risk measures from different location to each other and/or to some portfolio benchmark, c) suggesting a set of preventive policies that can be applied to the locations having the risk measures that significantly deviate from the benchmark.

One of the conventional ways to measure any risk is a prevalence rate which is the amount of losses related to the risk accidents over the total amount in a relevant portfolio. An alternative option of calculating loss rates is to use a number of risk events to the total number of events in a portfolio. For example, an operational loss rate associated with fraudulent transactions can be estimated as a ratio of the fraudulent transactions losses (or alternatively, a number of the fraudulent transactions) over the portfolio overall transaction volume (or total number of the portfolio transactions).

Methodology

A spatial analysis system is based on building funnel plots at several meaningful levels. A risk level funnel plot is originated from the normal approximation for a sampling distribution of a ratio (D. Anderson et al (2005)). A single funnel plot is a chart combining observed sample rates in the areas with the line connecting confidence intervals for the population rate estimate which are estimated for all possible sample sizes at the same confidence level. The funnel plots have proven to be effective and statistically valid tool for comparing the rates in small samples and often used in epidemiology for monitoring outbreaks of the contagious diseases (D. Dover (2010)).

In banking industry the funnel plots can be applied to the borrowers' default rates (i.e., 30-day, 60-day, and 90-day delinquency rates) in consumer credit portfolios and aimed at detection of areas with either significantly high or low deviation of the observed delinquency rates from the expected delinquency rate given a portfolio size. In practise, a risk analyst faces two issues: a) comparing the default rates from the areas with different portfolio sizes (also called loan volumes), b) monitoring concentration of defaults in the neighbouring areas.

The application of the funnel plots is straightforward in consumer credit portfolios, since these portfolios are quite homogeneous. These portfolios usually consist of standardized loan products sold to many borrowers that are relatively identical in their wealth. In contrast, application of the method commercial portfolios has strong reservations. The commercial portfolios are more heterogeneous and more concentrated, since they often consist of the loans that are customized and sold to businesses of different size, specialization, and areas of operation.

Algorithm of application the methodology

Suggested approach consists of six key elements below is designed to aggregate and effectively present information on loss rates.

1. Analyzing an overall portfolio area and dividing it into smaller geographic units with relatively homogeneous portfolios of sufficient size. Following statistic rule of thumb, it is recommended to have at least 30 objects (e.g., borrowers, transactions) in the unit portfolio.
2. Constructing four funnel plots at 80% and 90% confidence levels by estimating the approximate confidence intervals (D. Anderson and et al (2005)) for the benchmark loss rate ($\hat{p}$) for all observed sample sizes (n_i):

$$\hat{p} = \frac{1}{K} \sum_{i=1}^K \frac{D_i}{n_i}$$

$$I_{[0,1]} \left(\hat{p} \pm Z_{(1-\alpha/2)} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right) \quad (1.1)$$

$$n \in [\min(n_i), \max(n_i)]$$

where

$\hat{p}$ is a benchmark loss rate which is usually an overall portfolio loss rate estimated by an average rate in all areas of interest. However, some target value that is the same for all geographic units could be an alternative option for the benchmark loss rate;

D_i is a number of the loss events in all areas of interest i , $i = 1, \dots, K$;

$Z_{(1-\alpha/2)}$ is a value of the cumulative normal distribution that corresponds to $(1 - \alpha/2)\%$ probability. In 80% and 90% confidence intervals α equals 20% and 10%, correspondingly.

$I_{[0,1]}$ is an indicator function that limits the approximate confidence interval boundaries within a meaningful range.

3. Develop the funnel the plots by exhibiting all loss rates and their confidence intervals exhibited on y-axis and all potential unit sizes (n_i) on x-axis. At this chart a reference line is to be drawn to mark an overall portfolio loss rate.

The funnel plots will outline the boundaries of five risk groups: a) an extremely high risk group containing geographic units with their default rates above 95% funnel plot's zone; b) a moderately high risk group with the units having the default rates in 90-95% zone; c) normal risk group containing the geographic units with default rates in 10-90% zone; d) a moderately low risk group combining the units having their default rates in 5-10% zone; c) extremely low risk group with the geographic units' default rates in 0-5% zone.

4. Transferring the classification information onto geographic maps with assigning signal colours corresponding to the units in specific risk groups.

5. Analysis of the loss rates mapping that is focused on a) clustering of the units of the same or close risk groups, and b) consistency of their classification over the last 3-5 time periods;

6. Suggesting an effective policy to the units' portfolios that have the loss rates significantly different from the overall portfolio benchmark.

Application of signaling colours in mapping observations and suggested policy for risk groups

The Table 1 below suggest the signal colors and policies that can be applied to the units' portfolios that are consistently classified in the risk category for several time periods in a row. The policies below are developed for a retail loan portfolio to describe as an example.

Table 1. An application of signal colors and suggested policies

Risk category	Statistic confidence interval boundaries outlined by the funnel plots	Color	Suggested policy for a retail loan portfolio
Extremely high	95% and higher	Red	Emergency intervention: stop lending in the area, investigation of the situation
Moderate High	90-95%	Pink	Toughening the credit policy: higher credit limits and down-payment sizes; higher customers' risk score requirements
Normal risk	10-90%	Green	No change in credit policy, keep monitoring
Moderate Low	5%-10%	Light blue	Applying mild expansion credit policy: lower credit limits, down payment sizes; lower customers' risk score requirements.
Extremely Low	5% and lower	Blue	Applying aggressive expansion credit policy: significantly lower credit limits, down payment sizes, lower customers' risk score requirements

An application of the methodology to RML delinquency rates

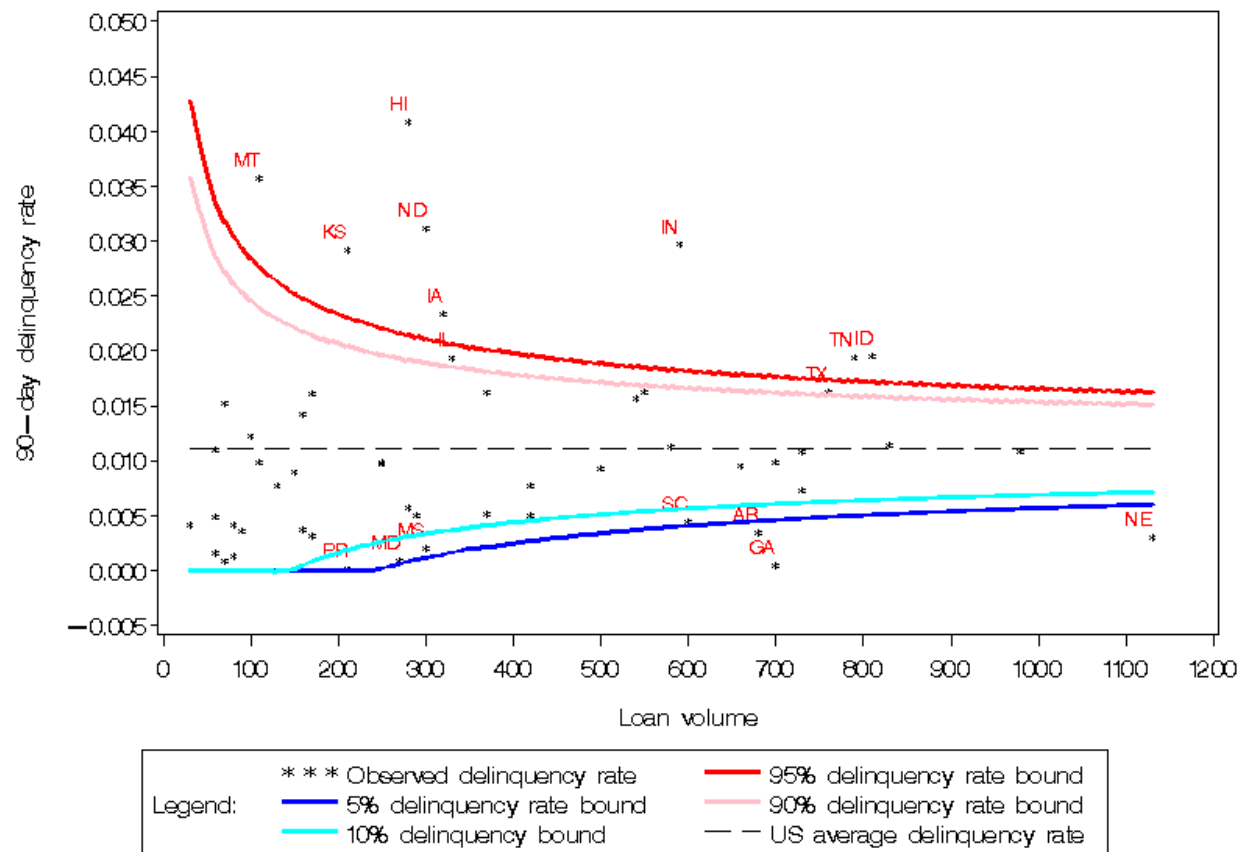
To demonstrate capabilities of the suggested approach, it is applied a 90-day delinquency rate in conventional residential mortgage loan (RML) portfolio. For this purpose, the delinquency rate and portfolio sizes are randomly generated for the United States ($i = 1, \dots, 52$ entities) with the following assumptions on distribution of the variables:

90-day delinquency rate, $\sim \text{Beta} (\alpha=0.98, \beta=97.02)$

Loan volume, $\sim 10 * \text{Negative Binomial} (p=0.05, k=2)$

The results produced by the approach for one of the simulations are summarized in the charts and tables below¹.

Chart 1. Funnel plot for 90—day delinquency rate in RML portfolio



¹ This application is developed and realized using SAS statistical software. The corresponding SAS code can be provided by an author on request.

Table 2. The portfolios with high delinquency rate

State name abbreviation	State Name	Loan volume	Observed delinquency rate	90% delinquency rate bound
ID	Idaho	810	0.01957	0.01586
TN	Tennessee	790	0.01944	0.01592
TX	Texas	760	0.01632	0.01602
IN	Indiana	590	0.02970	0.01668
IL	Illinois	330	0.01934	0.01854
IA	Iowa	320	0.02340	0.01866
ND	North Dakota	300	0.03114	0.01890
HI	Hawaii	280	0.04087	0.01918
KS	Kansas	210	0.02918	0.02042
MT	Montana	110	0.03572	0.02396

References

1. Enterprise Risk Management Committee (May 2003). Overview of Enterprise Risk Management. Casualty Actuarial Society Retrieved 2008-09-15.
2. David R. Anderson, Dennis J. Sweeney, Thomas A. Williams. Statistics for Business and Economics. 9th edition. Thomson South-Western, 2005
3. Doug Dover. Funnel Plots: Visualizing small area estimates. Presentation at SAS users group meeting in Edmonton, Canada: <http://www.sas.com/offices/NA/canada/downloads/presentations/Edmonton2010/Funnel.pdf>

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